EXACTNESS PROPERTIES OF THE PUSHFORWARD ALONG A POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The canonical split surjection from the stalk of the pushforward. Same chapter, Lemma lemma-stalk-skyscraper. Provides the split surjection which gives faithfulness, reflection of injections, surjections and isomorphisms.
- Every point comes from a point of the site. Same chapter, Lemma lemma-point-site-topos. Allows the pushforward to be computed by the formula for points coming from the site.
- The description of the pushforward along a point. Same chapter, Equation equation-skyscraper and Definition definition-pushforward-point. Supplies the formula from which the remaining properties follow.

Source reference. Original lemma 00YA: sources/stacks-sites/pinned-original-source.tex, source lines 8358–8358. Navigation pointer invoked at proof line 8358; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 04EI: sources/stacks-sites/pinned-original-source.tex, source lines 8377–8377. Navigation pointer invoked at proof line 8377; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original definition 00Y9: sources/stacks-sites/pinned-original-source.tex, source lines 8437–8437. Navigation pointer invoked at proof line 8437; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $p : Sh(pt) \to Sh(\mathcal{C})$ be a point of the topos associated to $\mathcal{C}$. The functor $p_* : Sets \to Sh(\mathcal{C})$ has the following properties: It commutes with arbitrary limits, it is left exact, it is faithful, it transforms surjections into surjections, it commutes with coequalizers, it reflects injections, it reflects surjections, and it reflects isomorphisms.*

Proof. Because p_* is a right adjoint it commutes with arbitrary limits and it is left exact. The fact that $p^{-1}p_*E \to E$ is a canonically split surjection implies that p_* is faithful, reflects injections, reflects surjections, and reflects isomorphisms. By Lemma 00YA we may assume that p comes from a point $u : \mathcal{C} \to Sets$ of the underlying site $\mathcal{C}$. In this case the sheaf p_*E is given by

$$p_*E(U) = \mathrm{Mor}_{Sets}(u(U), E)$$

see Equation (04EI) and Definition 00Y9. It follows immediately from this formula that p_* transforms surjections into surjections and coequalizers into coequalizers. □